

Scale-graded raw direct transfer, signed future last insertion, and reveal reanchoring boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope; authority

R-137 proved that the complete raw endpoint has two chronological triangles: the direct cells $t(k) \leq r$ and the future-feedback cells $t(k) > r$. It did not prove the spatial estimate for either triangle. This note separates the two remaining mechanisms and obtains the strongest statement currently justified by the accepted authorities.

First, absence of target heat is not by itself an obstruction to the direct Cartan estimate. On a scale-aligned cell whose complete raw coefficient path is Q_r -supported, the R-087 order-two paracomposition proof applies before heat. Target heat only contracts coefficient derivative bounds already available from the positive floor. The result gives the complete far-output six-row current constant $3/(40P)$, including the outer factor two in the R-084 raw OU identity. This is a conditional direct-current theorem. A new high-prefix fixture proves that chronological precedence alone does not imply the required spatial grade.

Second, finite Fubini does transpose the full R-079 future current–trace energy to a signed last-insertion telescope on each fixed ordered chart. It does not produce nonnegative R-135 atoms. At reveal weight γ , moving a future cell from its reveal root r to its insertion shell k creates the exact factor $2^{2\gamma(k-r)}$. At $s = 2/3$, $\gamma = 7/12$, and $k - r = 6$, the factor is 128. The wedge $r + 5 \leq m < k + 5$ is also near or balanced relative to the insertion and is not covered by a last-insertion FAR estimate. Thus bare finite Fubini plus an unweighted $\sum_k q_k$ cannot invoke R-135 on the future triangle.

The binding authorities are R-063, R-079, R-084, R-087, R-088, R-102, R-104, R-123, R-125, R-129, R-134, R-135, R-136, and R-137. R-138 claims novelty only for the scale-graded raw direct transfer, the complete signed current–trace last-insertion identity, the reveal-to-insertion reanchoring formula, and the resulting sharp ledger and collar criteria. It proves no complete production owner intertwiner, no production one-use charge sum, no signed forest-current lower form, no positive collar headroom, neither A13 gate, no Nelson theorem, and no Sector-A closure.

2. Index and ownership discipline

Fix one finite temporally faithful chart. Same-root visits are first recombined, equal-time insertions are grouped into one block, and every contraction-connected output is placed in an exhaustive cluster. Write o for a complete owner, $\ell = \tau(o)$ for its retained covariance root, $r > \ell$ for a reveal root, e for a chronological insertion event, $k = \kappa(e)$ for its physical spatial shell, and m for an output shell. On the aligned whole-shell chart $t(k) = k$.

For the six-row frame, retain

$$q_A(w) = \frac{w^T S_A w}{|w|^2 + e}, \quad F_A(w) = q_A(w)w, \quad \alpha_{\text{PF}} = \frac{5}{9}, \quad (2.1)$$

$$P_A(w; z) = 2(S_A w)^T z, \quad L_A(w; z) = 2\{S_A w - \alpha_{\text{PF}} F_A(w)\}^T z, \quad (2.2)$$

$$c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad P = 4 + 10^{-12}. \quad (2.3)$$

The unaveraged current is $J = C_6(W)V$ and $B = C_6(W)^* C_6(W)$. Current, trace, and R-063 forest are related owner coordinates; the forest is not an additional reserve after the trace has already been expanded into it.

3. Scale-graded raw direct spatial transfer

Fix $k \leq r$ on the aligned chart and put

$$u_{r,k,\theta} = B_{r,k} + \theta a_k, \quad 0 \leq \theta \leq 1. \quad (3.1)$$

Assume the following direct-cell data.

1. The complete paths $u_{r,k,\theta}$, the insertion a_k , and every fresh-root direction v_h are Q_r -supported, while a_k is Q_k -supported.
2. Strict past gives $D_r a_k = 0$, $D_r B_{r,k} = v_h$, and

$$\sup_x \sum_h |v_h(x)|^2 \leq \kappa_0 2^{-r}, \quad \sup_x \sum_h |Dv_h(x)|^2 \leq \kappa_1 2^r. \quad (3.2)$$

3. The positive-floor maps F_A have the R-087 derivative bounds through order five. The relevant conditional expectation, root OU operator, and deterministic spatial projection are contractions and commute on the displayed smooth-cylinder product.
4. The complete owner intertwiner identifies $D_r P_r Y_{o,k}$ with the conditional raw three-channel current atom. This is an explicit hypothesis here, not a conclusion silently imported from R-137.

The scalar Cartan atom is

$$\begin{aligned} \mathfrak{t}_{A,r,k}^{\text{raw}}[v_h] &= \int_0^1 \{ D^2 F_A(u)[a_k, v_h]^T Du + D F_A(u)[v_h]^T D a_k \\ &\quad + D F_A(u)[a_k]^T D v_h \} d\theta. \end{aligned} \quad (3.3)$$

Equivalently, (3.3) is the fundamental-theorem-of-calculus form of R-137 (6.1). No future control is frozen in obtaining this identity.

For $1/3 < \rho < 1/2$ and $1/2 < s < 3\rho - 1/2$, define

$$W_k^{\text{raw}} := \sup_{r \geq k} \sum_A \mathbb{E} \int_0^1 (1 + \|u_{r,k,\theta}\|_{C^\rho})^6 \times (\|a_k\|_2^2 + \|D a_k\|_2^2) d\theta, \quad (3.4)$$

$$q_k^{\text{Cart}} := C_e 2^{-(6\rho-1)k} W_k^{\text{raw}}. \quad (3.5)$$

Theorem 3.1 (scale-graded raw direct Cartan estimate)

Under the preceding hypotheses, for every $m \geq r + 5$,

$$\boxed{\sum_A \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(r)} P_r \mathfrak{t}_{A,r,k}^{\text{raw}}\|_{\text{HS}}^2 dt \leq 2^{-2s(m-k)} q_k^{\text{Cart}}.} \quad (3.6)$$

Proof. The positive floor gives the derivative bounds used in R-087 before any target heat is applied. Apply the identical order-two parilinearisation to (3.3). Every principal term is supported in Q_{r+4} and is therefore killed by Π_m for $m \geq r+5$. The three remainders have the R-087 powers

$$2^{-4\rho m}2^{(1-2\rho)r}, \quad 2^{-6\rho m}2^{-r}, \quad 2^{-6\rho m}2^r. \quad (3.7)$$

Root summation uses (3.2), conditional expectation and root OU are L^2 contractions, and $r \geq k$ converts the remaining scale to (3.5). The excess exponents are positive precisely in the stated range. $\square$

4. Complete six-row constant

The P_A rows and the $2S_A w$ part of L_A are linear in w . Their second insertion/root variation vanishes, and their remaining principal far projection is killed under the support hypotheses. Only $-2\alpha_{\text{PF}}\sqrt{c_1}F_A$ survives. Including the leading factor two in the raw R-084 OU identity gives

$$8\alpha_{\text{PF}}^2 c_1 = 8 \left(\frac{5}{9}\right)^2 \frac{243}{8000P} = \frac{3}{40P}. \quad (4.1)$$

Consequently (3.6) yields

$$2 \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(r)} P_r \mathfrak{t}_{r,k}^{C_6, \text{raw}}\|_{\text{HS}}^2 dt \leq 2^{-2s(m-k)} q_k^{C_6}, \quad q_k^{C_6} = \frac{3}{40P} q_k^{\text{Cart}}. \quad (4.2)$$

In R-137's convention, which first doubles the scalar Cartan charge, the same statement is $q_k^{C_6} = (3/(80P))(2q_k^{\text{Cart}})$.

At $\rho = 2/5$ and $s = 2/3$,

$$6\rho - 1 = \frac{7}{5}, \quad 6\rho - 1 - 2s = \frac{1}{15}, \quad 4\rho - 2s = \frac{4}{15}, \quad 6\rho - 2s = \frac{16}{15}. \quad (4.3)$$

Thus the direct far exponent is $2^{-4(m-k)/3}$. Equation (4.2) remains conditional on the complete-owner and scale-grade hypotheses; it does not prove their production realization or $\sum_k q_k^{C_6} < \infty$.

5. Chronology does not imply spatial grade

The Q_r prefix hypothesis cannot be replaced by $t(k) \leq r$. On one spatial circle take

$$F_e(x) = \frac{x^3}{x^2 + e}, \quad b = \sqrt{e}, \quad F_e''(b) = \frac{1}{2\sqrt{e}} \neq 0. \quad (5.1)$$

For $N = 2^M$ and $\rho = 2/5$, let

$$B_N(\xi, x) = b + \xi + \delta N^{-\rho} \cos(Nx), \quad v = \partial_\xi B_N = 1, \quad a = a_0, \quad (5.2)$$

where a_0 is a sufficiently small nonzero constant. Then $Da = Dv = 0$ and the exact first channel is

$$T_N = \{F_e'(B_N + a) - F_e'(B_N)\} DB_N. \quad (5.3)$$

On a fixed positive-probability neighbourhood of $\xi = 0$, Taylor's theorem and (5.1) give, for sufficiently small fixed a_0, δ ,

$$\mathbb{E} \|\Pi_M T_N\|_2^2 \geq c_e a_0^2 \delta^2 N^{2-2\rho}. \quad (5.4)$$

Meanwhile $\|B_N\|_{C^\rho}$ has moments uniform in N , so the low- k R-087-form charge is $O(a_0^2)$. At $s = 2/3$, the proposed right side is $O(a_0^2 N^{-4/3})$, and the ratio grows like

$$N^{2-2\rho+2s} = N^{38/15}. \quad (5.5)$$

This establishes NG-2026-07-31-A13-CHRONOLOGY-ONLY-SPATIAL-GRADE. It is a proof-method fixture, not a full production-action counterexample. It shows that the direct theorem needs either an actual Q_r -supported raw prefix or a unique maximal-spatial-grade reassignment before any charge is declared.

6. Exact signed future last-insertion telescope

Let $e = 1, \dots, N$ be the ordered insertion events and let t_e be the chronological time of event e . Write

$$J^{(e)}, \quad B^{(e)} = C_6(W^{(e)})^* C_6(W^{(e)}), \quad \delta_e J = J^{(e)} - J^{(e-1)}, \quad \delta_e B = B^{(e)} - B^{(e-1)}. \quad (6.1)$$

Let $n(r)$ be the last event with $t_e \leq r$. The complete R-079 current–trace future block at reveal r is

$$\begin{aligned} \mathfrak{J}_r &= \mathbb{E} \langle d_r J^{(n(r))}, d_r (J^{(N)} - J^{(n(r))}) \rangle \\ &\quad + \frac{1}{2} \mathbb{E} \|d_r (J^{(N)} - J^{(n(r))})\|^2 - \frac{1}{2} \mathbb{E} \int (B^{(N)} - B^{(n(r))}) : \Delta \Gamma_r. \end{aligned} \quad (6.2)$$

Theorem 6.1 (complete signed last-insertion transpose)

For every fixed finite chart for which the displayed current terms lie in L^2 and the trace pairings lie in L^1 ,

$$\boxed{\sum_{r>\ell} \mathfrak{J}_r = \sum_e \sum_{\ell < r < t_e} \left[\mathbb{E} \operatorname{Re} \langle d_r J^{(e-1)}, d_r \delta_e J \rangle + \frac{1}{2} \mathbb{E} \|d_r \delta_e J\|^2 - \frac{1}{2} \mathbb{E} \int \delta_e B : \Delta \Gamma_r \right]}. \quad (6.3)$$

Equivalently, with

$$Q_e = P_{t_e-1} - P_\ell, \quad \Gamma_e^- := \sum_{\ell < r < t_e} \Delta \Gamma_r, \quad (6.4)$$

the insertion- e owner is

$$\boxed{\mathfrak{J}_e^{\text{LI}} = \mathbb{E} \operatorname{Re} \langle Q_e J^{(e-1)}, Q_e \delta_e J \rangle + \frac{1}{2} \mathbb{E} \|Q_e \delta_e J\|^2 - \frac{1}{2} \mathbb{E} \int \delta_e B : \Gamma_e^-,} \quad (6.5)$$

and $\sum_{r>\ell} \mathfrak{J}_r = \sum_e \mathfrak{J}_e^{\text{LI}}$.

Proof. For fixed r , telescope the cumulative current energy using

$$\frac{1}{2} (\|x + y\|^2 - \|x\|^2) = \operatorname{Re} \langle x, y \rangle + \frac{1}{2} \|y\|^2. \quad (6.6)$$

Telescope $B^{(N)} - B^{(n(r))}$ in the same insertion order. This gives the summand in (6.3). All sums are finite, so exchange the r, e order. Finally, martingale orthogonality identifies the sums over r with the two Q_e inner products, and covariance additivity gives Γ_e^- . $\square$

The coordinate is canonical only under the convention that event e owns the whole cumulative-energy difference from endpoint $e - 1$ to endpoint e . It is signed and chart-dependent. Scalar increments $1, -1$ have owners $1/2, -1/2$ and terminal half-energy zero; replacing them by separate positive squares gives one and destroys cancellation. Increments $1, 1$ have owners $1/2, 3/2$ and terminal half-energy two. R-104 subdivision may redistribute these owners while preserving their recombined total.

7. Reveal reanchoring and the uncovered future wedge

On the aligned chart a future cell has $r < k$. Suppose, only for the far relative-to-insertion part, that a nonnegative cell estimate were proved:

$$\mathbb{E} \|\Pi_m T_{r,k}\|^2 \leq 2^{-2s(m-k)} q_{r,k}, \quad m \geq k + 5. \quad (7.1)$$

Multiplication by the R-135 reveal-anchored weight gives the exact identity

$$\boxed{2^{2\gamma(m-r-5)} 2^{-2s(m-k)} = 2^{-10\gamma} 2^{2\gamma(k-r)} 2^{-2(s-\gamma)(m-k)}. \quad (7.2)$$

The new factor grows with causal depth. At $s = 2/3$, $\gamma = 7/12$, and $k - r = 6$, it is

$$2^{2\gamma(k-r)} = 2^7 = 128. \quad (7.3)$$

Thus the old condition $\sum_k q_k \leq Q$ does not by itself control the future reveal-weighted ledger. A sufficient causal charge is

$$\sum_k \sum_{r < k} 2^{2\gamma(k-r)} q_{r,k} \leq Q_{\text{fut}}, \quad (7.4)$$

or the stronger pointwise factorisation

$$q_{r,k} \leq 2^{-2\gamma(k-r)} \bar{q}_k, \quad \sum_k \bar{q}_k \leq Q_{\text{fut}}. \quad (7.5)$$

Moreover, for $r + 5 \leq m < k + 5$, the output is FAR relative to the reveal but not relative to the last insertion. High-high-to-low leakage prevents this wedge from being declared empty. It must remain in the complete signed near/balanced current-trace-forest owner or receive a separate production estimate. Equations (7.2)–(7.5) establish NG-2026-07-31-A13-BARE-LAST-INSERTION-R135-REANCHORING: finite Fubini, $m \geq k + 5$, and an unweighted $\sum_k q_k$ are insufficient for the reveal-weighted future ledger. This is an inference no-go, not a production counterexample and not a rejection of a gap-weighted or direct signed route.

8. Sharp scalar charge criteria

After complete same- k recombination, let $a_{r,m,k}^{\text{dir}}$ be the nonnegative left side of the legally centred direct atom estimate in the R-135 convention without the R-137 outer factor two. Define

$$q_k^{*,\text{dir}}(s) := \sup_{r \geq k, m \geq r+5} 2^{2s(m-k)} a_{r,m,k}^{\text{dir}}. \quad (8.1)$$

Then deterministic nonnegative charges satisfying the direct inequalities and $\sum_k q_k \leq Q$ exist if and only if

$$\boxed{\sum_k q_k^{*,\text{dir}}(s) \leq Q.} \quad (8.2)$$

The minimal scalar ledger is $q_k = q_k^{*,\text{dir}}$. In the raw OU convention the charges and total are doubled.

For a separated future FAR route define

$$q_{r,k}^{*,\text{fut}}(s) := \sup_{m \geq k+5} 2^{2s(m-k)} a_{r,m,k}^{\text{fut}}. \quad (8.3)$$

The exact reanchored scalar cost is

$$\boxed{\mathfrak{C}_{s,\gamma}^{\text{fut}} := \sum_k \sum_{r < k} 2^{2\gamma(k-r)} q_{r,k}^{*,\text{fut}}(s).} \quad (8.4)$$

This is the minimal cellwise nonnegative cost for that separated future FAR strategy. It is stronger than a direct estimate of the signed owner (6.5), which may retain cancellations. A matrix diagonal-majorant/Bessel theorem is stronger still and is not required by the R-135 weighted-Cauchy argument.

The open theorem is therefore not the choice of symbols called q_k . It is a cutoff-, chart-, control-, visit-, cluster-, and refinement-uniform production estimate for (8.2), or for (8.4) plus the uncovered signed wedge, with source and terminal sextic budgets used once.

9. Conditional finite-collar consequence

The R-134 direct fixed-collar constant is

$$D_{7/12} = \frac{2^{-35/6}}{(1 - 2^{-7/12})^2(1 - 2^{-7/6})} = 0.285928885855479 \dots \quad (9.1)$$

Suppose a compatible complete charge Q has actually been proved and let Λ_{own} be the certified owner-conversion factor. If

$$h_C := a_* - a_0(C) > 0, \quad a_* = 2\sqrt{(e_0 - \sigma_0)(f_0 - \sigma_0)}, \quad \sigma_0 = \frac{\kappa_{\text{low}}^2}{d}, \quad (9.2)$$

then the exact fixed-collar acceptance condition is equivalent to

$$Q < \frac{h_C^2}{\Lambda_{\text{own}}^2 D_{7/12}} 2^{7(C-5)/6}. \quad (9.3)$$

For the raw OU convention, substitute $Q_{\text{raw}}/2$ for Q . Equation (9.3) does not prove that h_C is positive. The production values of Q , Λ_{own} , e_0 , f_0 , d , κ_{low} , and $a_0(C)$ remain unproved, so no collar is certified here.

10. Proof, failure, and successor map

Route	Verdict	Reusable conclusion
Raw direct paracomposition	proved conditionally	Target heat is unnecessary once the complete path is Q_r -supported and the owner/commutation hypotheses hold.
Six-row far constant	proved conditionally	The nonlinear row coefficient including raw OU is exactly $3/(40P)$.
Chronology as spatial support	rejected	A bounded- $\mathcal{C}^{2/5}$ high prefix violates the claimed low- k decay by $N^{38/15}$.
Future finite Fubini	proved exactly	The complete current-trace future block has the signed owner (6.5) on each fixed chart.
Positive-square last insertion	rejected	Increments $1, -1$ cancel only in the signed cumulative-energy coordinate.
Bare R-135 reindexing	rejected	Reveal reanchoring costs $2^{2\gamma(k-r)}$ and leaves the future near wedge.
Sharp scalar charges	exact criterion	Equations (8.2) and (8.4) identify the actual direct and separated-future burdens.
Production one-use	open	Complete owner realization, charge summability, signed wedge, and headroom remain.

The proof-search records are EXP-000546–EXP-000551. They preserve the direct advance, both new method failures, the signed future transpose, the sharp charge/collar criterion, and the adversarial successor audit. The two new negative authorities are the chronology-only spatial-grade no-go and the bare last-insertion R-135 reanchoring no-go.

The corrected successor remains

$$\boxed{\text{A13-CLASSII-COMPLETE-RAW-DOOB-DIRECT-FUTURE-SPATIAL-ONE-USE-BOUND.}} \quad (10.1)$$

It must prove the production scale-grade/owner intertwiner for the direct branch, and for the future branch either prove the gap-weighted charge (8.4) together with the signed near wedge or estimate the whole signed owner (6.5) directly. It must then prove one deterministic once-owned total, retain the R-063 forest and low objects once, and test (9.3) with genuinely positive headroom.

11. Devil's-advocate review

1. **Objection: R-135 already proves Theorem 3.1. VALID provenance correction.** R-135 owns the post-heat sequential atom theorem. R-138 proves that the same paracomposition works before target heat under explicit raw support, owner, and commutation hypotheses; it does not reclaim R-135 or prove those hypotheses in production.
2. **Objection: every chronological prefix is automatically Q_r -supported. UPHELD as false.** The high deterministic prefix (5.2) is earlier in time but remains at spatial shell M . Temporal order and Fourier grade are different data.
3. **Objection: linear six-row pieces contribute another far constant. DISMISSED under the theorem hypotheses.** Their mixed second variation vanishes and the remaining principal products lie in Q_{r+4} . The only far nonlinear coefficient is $-2\alpha_{PF}\sqrt{c_1}F_A$.
4. **Objection: finite Fubini makes each future owner nonnegative. UPHELD as false.** Formula (6.6) assigns a signed cross plus square. The 1, -1 fixture has owners $1/2, -1/2$.
5. **Objection: R-063 forest may be added after (6.5) as a fresh reserve. UPHELD as duplicate ownership.** The trace term in (6.5) and its adapted R-063 forest expansion are alternative coordinates of the same owner.
6. **Objection: replacing r by k in the output weight is only notation. UPHELD as false.** Equation (7.2) gives the exact causal factor and (7.3) gives 128 at gap six.
7. **Objection: the future wedge $r + 5 \leq m < k + 5$ is empty by support. UPHELD as unproved.** High-high-to-low interactions can occupy it; only a complete signed near/balanced theorem can remove or pay it.
8. **Objection: the sharp charge definition proves its own uniform sum. UPHELD as circular.** Equations (8.1)–(8.4) identify minimal costs; their production uniformity is exactly the remaining analytic theorem.
9. **Objection: increasing the collar certifies headroom. UPHELD as false without separate data.** Equation (9.3) requires $h_C > 0$ and a proved total charge. Neither is created by tail decay.

External review is invited for the support radius $r + 4$, the six-row factor $3/(40P)$, the lower-bound scope of (5.4), the current-trace signs and halves in (6.3)–(6.5), the reanchoring factor in (7.2), the uncovered future wedge, the raw OU factor two, and the no-overclaim boundary below.

12. Reproduction and Result footer

Run from the repository root:

```
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
& $py codes/foundations/
  a13_classii_scale_graded_direct_future_reanchoring_boundary.py
& $py codes/foundations/
  a13_classii_scale_graded_direct_future_reanchoring_boundary_independent.py
& $py codes/foundations/
  a13_classii_scale_graded_direct_future_reanchoring_boundary_verify.py
```

Result ID: A13-CLASSII-SCALE-GRADED-DIRECT-FUTURE-REANCHORING-
BOUNDARY

Ledger ID: R-138

Precise statement: Under an explicit Q_r -supported raw path, production derivative bounds, legal commutations, and a complete-owner intertwiner, the R-087 paracomposition proves the raw direct six-row far estimate with coefficient $3/(40P)$. On each fixed finite chart, the complete R-079 future current--trace energy has an exact signed last-insertion telescope. Reveal reanchoring costs $2^{(2\gamma(k-r))}$ and leaves a future near wedge, so bare finite Fubini plus an unweighted q sum cannot invoke R-135.

Scope: fixed finite cutoff and positive floor; bounded smooth cylindrical data; aligned direct cells only under the explicit spatial-grade theorem; fixed ordered charts for the signed future identity; same-root visits, equal-time events, and exhaustive clusters recombined first.

Dependencies: R-063, R-079, R-084, R-087, R-088, R-102, R-104, R-123, R-125, R-129, R-134, R-135, R-136, and R-137.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/
a13_classii_scale_graded_direct_future_reanchoring_boundary_verify.py

Expected output: R-138 integrated PASS with primary and independent children, deterministic PDF, all-page visual QA, and release evidence.

Falsification gate: failure of the direct exponents, six-row factor, F_e second derivative, obstruction exponent, signed energy/trace telescope, cancellation fixtures, reanchoring identity or factor 128, scalar charge criterion, collar threshold, any authority/hash/PDF/exploration/negative pin, or any no-overclaim scope assertion.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no complete production owner intertwiner, chronology-spatial grade theorem over the directed union, production direct/future q sum, future near/balanced signed lower form, positive collar headroom, low or matching data, strict gap, absolute anchor, A13 gate, OVERLAP_src, Nelson theorem, removal, interacting measure, or Sector-A closure is proved.

Next required action: prove the complete production scale-grade/owner intertwiner and either the gap-weighted future ledger plus signed near wedge or a direct estimate of the whole signed last-insertion owner; then prove one deterministic once-owned charge sum and certify finite-collar headroom.

Proof complete: false. Sector A closed: false.